

1. Write the program to read the content of one.txt file and write the content to the two.txt file using `BufferInputStream` and `BufferOutputStream`.

```
import java.io.*;

public class workshop8 {
    Run | Debug
    public static void main(String[] args) {
        String inputFile = "one.txt";
        String outputFile = "two.txt";

        try (
            BufferedInputStream inputStream = new BufferedInputStream(new FileInputStream(inputFile));
            BufferedOutputStream outputStream = new BufferedOutputStream(new FileOutputStream(outputFile));
        ) {
            byte[] buffer = new byte[1024];
            int bytesRead;
            while ((bytesRead = inputStream.read(buffer)) != -1) {
                outputStream.write(buffer, 0, bytesRead);
            }
            System.out.println("File copied successfully!");
        } catch (IOException e) {
            System.err.println("Error copying file: " + e.getMessage());
        }
    }
}
```

1 error

```
PS C:\Users\sigma\OneDrive\Desktop\java> javac workshop8.java
```

```
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8
```

```
File copied successfully!
```

```
PS C:\Users\sigma\OneDrive\Desktop\java> 
```

2. Rewrite the program from Question no 5 using `BufferedReader` and `BufferWriter`.

```
static class Student {  
    private String firstName;  
    private String lastName;  
    private String address;  
    private String contactNumber;  
    private String semester;  
  
    public Student(String firstName, String lastName, String address, String contactNumber, String semester) {  
        this.firstName = firstName;  
        this.lastName = lastName;  
        this.address = address;  
        this.contactNumber = contactNumber;  
        this.semester = semester;  
    }  
  
    public String toCSVFormat() {  
        return String.format("%s,%s,%s,%s,%s", firstName, lastName, address, contactNumber, semester);  
    }  
}
```

3. You have the text file with the given number range from 1 to 10. Each number is separated by space in between. Write the program to read the number from the text file then write the odd number into odd.txt file and even number into even.txt file.

The program should have the following methods.

- a. `public List<Integer> getEven(List<Integer> numbers);`
- b. `public List<Integer> getOdd(List<Integer> numbers);`
- c. `public void write(List<Integer> numbers, String filePath);`
- d. `public void read(String filePath);`

```
import java.io.*;
import java.util.ArrayList;
import java.util.List;

public class Workshop8 {
    Run | Debug
    public static void main(String[] args) {
        String inputFile = "workshop8.txt";
        String evenOutputFile = "even.txt";
        String oddOutputFile = "odd.txt";

        Workshop8 workshop = new Workshop8();
        List<Integer> numbers = workshop.read(inputFile);

        List<Integer> evenNumbers = workshop.getEven(numbers);
        List<Integer> oddNumbers = workshop.getOdd(numbers);

        workshop.write(evenNumbers, evenOutputFile);
        workshop.write(oddNumbers, oddOutputFile);

        System.out.println("Even and odd numbers have been written to their respective files.");
    }

    public List<Integer> getEven(List<Integer> numbers) {
        List<Integer> evenNumbers = new ArrayList<>();
        for (int num : numbers) {
            if (num % 2 == 0) {
                evenNumbers.add(num);
            }
        }
        return evenNumbers;
    }
}
```

```
    public List<Integer> getOdd(List<Integer> numbers) {
        List<Integer> oddNumbers = new ArrayList<>();
        for (int num : numbers) {
            if (num % 2 != 0) {
                oddNumbers.add(num);
            }
        }
        return oddNumbers;
    }

    public void write(List<Integer> numbers, String filePath) {
        try (PrintWriter writer = new PrintWriter(filePath)) {
            for (int num : numbers) {
                writer.println(num);
            }
        } catch (FileNotFoundException e) {
            System.err.println("Error writing to file: " + e.getMessage());
        }
    }

    public List<Integer> read(String filePath) {
        List<Integer> numbers = new ArrayList<>();
        try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
            String line;
            while ((line = reader.readLine()) != null) {
                numbers.add(Integer.parseInt(line));
            }
        }
        return numbers;
    }
}
```

```
public List<Integer> read(String filePath) {  
    List<Integer> numbers = new ArrayList<>();  
    try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {  
        String line;  
        while ((line = reader.readLine()) != null) {  
            String[] tokens = line.split("\\s+");  
            for (String token : tokens) {  
                numbers.add(Integer.parseInt(token));  
            }  
        }  
    } catch (IOException e) {  
        System.err.println("Error reading file: " + e.getMessage());  
    }  
    return numbers;  
}
```

```
PS C:\Users\sigma\OneDrive\Desktop\java> javac workshop8.java  
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8  
File copied successfully!  
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8.java  
Even and odd numbers have been written to their respective files.  
PS C:\Users\sigma\OneDrive\Desktop\java>
```

4. Write a program that reads a text file and counts the number of words in it.
Display the total count at the end.

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class Workshop8 {
    Run | Debug
    public static void main(String[] args) {
        String filename = "workshop8.txt";
        int wordCount = countWords(filename);
        System.out.println("Total number of words in the file: " + wordCount);
    }

    public static int countWords(String filename) {
        int count = 0;
        try (BufferedReader reader = new BufferedReader(new FileReader(filename))) {
            String line;
            while ((line = reader.readLine()) != null) {
                String[] words = line.split("\\s+");
                count += words.length;
            }
        } catch (IOException e) {
            System.err.println("Error reading file: " + e.getMessage());
        }
        return count;
    }
}
```

```
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8.jsvs
Error: Could not find or load main class workshop8.jsvs
Caused by: java.lang.ClassNotFoundException: workshop8.jsvs
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8.java
Total number of words in the file: 3
PS C:\Users\sigma\OneDrive\Desktop\java> |
```

5. Write a program to take the student information from the user and write the content to the csv file. The student class should have the following attributes.
 - a. firstName, lastName, address, contactNumber, semester

```

import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class Workshop8 {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter student information:");
        System.out.print("First Name: ");
        String firstName = scanner.nextLine();

        System.out.print("Last Name: ");
        String lastName = scanner.nextLine();

        System.out.print("Address: ");
        String address = scanner.nextLine();

        System.out.print("Contact Number: ");
        String contactNumber = scanner.nextLine();

        System.out.print("Semester: ");
        String semester = scanner.nextLine();

        Student student = new Student(firstName, lastName, address, contactNumber, semester);

        writeStudentInfoToCSV(student);
    }
}

```

b.

```

        writeStudentInfoToCSV(student);

        System.out.println("Student information written to workshop8.csv successfully.");

        scanner.close();
    }

    public static void writeStudentInfoToCSV(Student student) {
        try (FileWriter writer = new FileWriter("workshop8.csv", true)) {
            writer.write(student.toCSVFormat() + "\n");
        } catch (IOException e) {
            System.err.println("Error writing to CSV file: " + e.getMessage());
        }
    }

    static class Student {
        private String firstName;
        private String lastName;
        private String address;
        private String contactNumber;
        private String semester;

        public Student(String firstName, String lastName, String address, String contactNumber, String semester) {
            this.firstName = firstName;
            this.lastName = lastName;
            this.address = address;
            this.contactNumber = contactNumber;
            this.semester = semester;
        }

        public String toCSVFormat() {

```

c.

Do you want to install the recommended extension from marketplace for

```
static class Student {  
    private String firstName;  
    private String lastName;  
    private String address;  
    private String contactNumber;  
    private String semester;  
  
    public Student(String firstName, String lastName, String address, String contactNumber, String semester) {  
        this.firstName = firstName;  
        this.lastName = lastName;  
        this.address = address;  
        this.contactNumber = contactNumber;  
        this.semester = semester;  
    }  
  
    public String toCSVFormat() {  
        return String.format("%s,%s,%s,%s,%s", firstName, lastName, address, contactNumber, semester);  
    }  
}
```

```
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8.java  
Total number of words in the file: 3  
PS C:\Users\sigma\OneDrive\Desktop\java> java workshop8.java  
Enter student information:  
First Name: Bibek  
Last Name: Bhandari  
Address: purano guheswori  
Contact Number: 9801692864  
Semester: 2
```

```
workshop8.csv  
Bibek,Bhandari,purano guheswori,9801692864,2
```

6. You have a text file which contains the information about the Electronic and Computing product separated by ; (semicolon). The text file has the following attributes.

a. product(String), price(Double)

Write a program to read the content from the text file, find the average sales of the product and export it to the csv file.

Make sure you have the multiple products in the text file. For example

product;price

Laptop; 40,000.0

SmartPhone; 50,000.0

SmartPhone; 60,000.0

Laptop;80,000

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.HashMap;
import java.util.Map;

public class workshop8 {
    Run | Debug
    public static void main(String[] args) {
        String inputFile = "workshop8.txt";
        String outputFile = "average_sales.csv";

        Map<String, Double> salesMap = readSalesFromFile(inputFile);
        if (!salesMap.isEmpty()) {
            calculateAndWriteAverageSales(salesMap, outputFile);
            System.out.println("Average sales exported to " + outputFile + " successfully.");
        } else {
            System.out.println("No sales data found in the input file.");
        }
    }

    public static Map<String, Double> readSalesFromFile(String filename) {
        Map<String, Double> salesMap = new HashMap<>();
        try (BufferedReader reader = new BufferedReader(new FileReader(filename))) {
            String line;
            while ((line = reader.readLine()) != null) {
                String[] parts = line.split(";");
                if (parts.length == 2) {
                    String product = parts[0];
                    double price = Double.parseDouble(parts[1]);
                    salesMap.put(product, price);
                }
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
        return salesMap;
    }

    public static void calculateAndWriteAverageSales(Map<String, Double> salesMap, String outputFile) {
        try (FileWriter writer = new FileWriter(outputFile)) {
            for (Map.Entry<String, Double> entry : salesMap.entrySet()) {
                writer.write(entry.getKey() + ";" + entry.getValue() + "\n");
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```



```

        double price = Double.parseDouble(parts[1].trim());
        if (salesMap.containsKey(product)) {
            salesMap.put(product, salesMap.get(product) + price);
        } else {
            salesMap.put(product, price);
        }
    }
}
} catch (IOException | NumberFormatException e) {
    System.err.println("Error reading file: " + e.getMessage());
}
return salesMap;
}

public static void calculateAndWriteAverageSales(Map<String, Double> salesMap, String outputFile) {
    try (FileWriter writer = new FileWriter(outputFile)) {
        writer.write("Product,Average Sales\n");
        for (Map.Entry<String, Double> entry : salesMap.entrySet()) {
            double averageSales = entry.getValue() / 2; // Assuming two entries for each product in the file
            writer.write(entry.getKey() + "," + averageSales + "\n");
        }
    } catch (IOException e) {
        System.err.println("Error writing to file: " + e.getMessage());
    }
}

```

7. With the reference of Question 5. Read the content from the csv file and export it to the text file. Here, the name of the text file should be the name of the header in the csv file and the content of the text file should be the value of the header.

For Example :-

CSV File

FIRST_NAME, LAST_NAME, ADDRESS, CONTACT_NUMBER, SEMESTER

John,Doe,123 Main St, Anytown USA,555-1234,Fall 2023

Jane,Smith,456 Oak Rd, Somecity NY,555-5678,Spring 2024

Michael,Johnson,789 Elm Blvd, Othertown CA,555-9012,Summer 2023

Emily,Williams,321 Maple Ave, Somewhere FL,555-3456,Fall 2024

David,Brown,654 Pine St, Anycity TX,555-7890,Spring 2023

FIRST_NAME.txt file

John

Jane

Michael

Emily

David

LAST_NAME.txt file

Doe

Smith

Johnson

Williams

Brown

and so on.

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.ArrayList;
import java.util.List;

public class Workshop8 {
    Run | Debug
    public static void main(String[] args) {
        String csvFile = "workshop8.csv";
        List<String> headers = new ArrayList<>();

        try (BufferedReader br = new BufferedReader(new FileReader(csvFile))) {
            String line;
            if ((line = br.readLine()) != null) {
                String[] headersArray = line.split(",");
                for (String header : headersArray) {
                    headers.add(header.trim());
                    createTextFile(header.trim());
                }
            }

            while ((line = br.readLine()) != null) {
                String[] values = line.split(",");
                for (int i = 0; i < headers.size(); i++) {
                    writeTextFile(headers.get(i).trim(), values[i].trim());
                }
            }

            System.out.println("Data exported to text files successfully.");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

```
public static void createTextFile(String header) {  
    try (FileWriter writer = new FileWriter(header + ".txt")) {  
        // Create empty file  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}  
  
public static void writeTextFile(String header, String value) {  
    try (FileWriter writer = new FileWriter(header + ".txt", true)) {  
        writer.write(value + "\n");  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}
```

Note: Make sure to create your own text file and content.